Solving Power Density and Bandwidth Challenges in AI Infrastructure

Executive Session

OCP Global Summit 2026

Gaurav Jain, Sr. Dir., Product Management

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EVERY CONNECTION COUNTS

Driving 800VDC through the Racks

Lower Current, Smaller Conductors, Lower Resistive Losses

800VDC CREATES NEW
CONNECTION REQUIREMENTS

- Touch Protection
- Creepage and Clearance
- Controlled Mating
- Interlocks
- De-energized Maintenance
- Connector Condition Monitoring

TE IS DRIVING 800V POWER

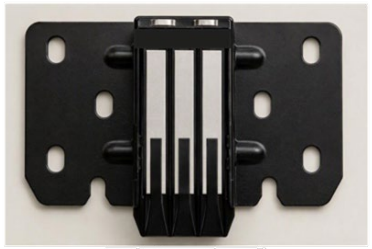
1MW AND BEYOND

Building blocks for next-generation 800V rack architectures

800V SIDECAR
High Power scalable options for Busbar and PSU

800V IT RACK BUSBAR
250kW to 1.6MW
scalable depth options

COMPACT MATING
Sense and precharge fit within 35mm busbar width



NOTE: The TE products shown and discussed in this presentation are shown for reference only and are not currently licensed contributions to any OCP specifications

LEARN MORE AT TE PRESENTATION
Enabling 1MW+ IT Rack Power with an Open Rack Busbar Interface

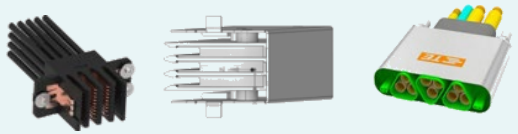
Thursday, Oct 15 I 10:15 AM
SJCC – Concourse Level, 230A

One Sidecar Architecture, Two Paths to the Compute Load

Sidecar Separates Power Conversion from Compute. Output Architecture Determines Where Final Conversion Occurs

PATH A

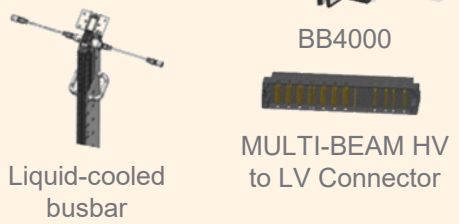
HIGH-VOLTAGE IT RACK



High Vol busbar connector | High Vol busbar | Rack-to-rack connections

PATH B

50V IT RACK

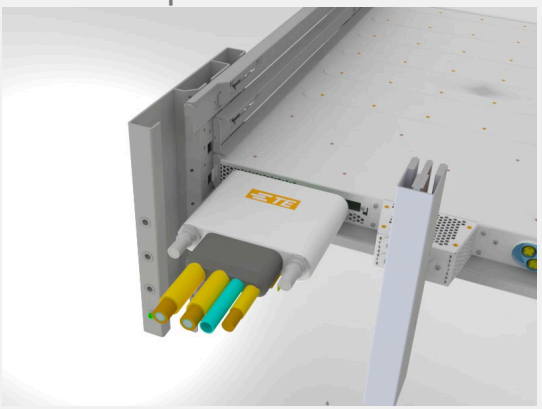


BB4000 | Liquid-cooled busbar | MULTI-BEAM HV to LV Connector

INDUSTRY-LEADING

1OU Rack-to-Rack Power Connection

Designed for air- and liquid-cooled 800V architecture



HIGHER CURRENT

More power per conductor cross-section

LIQUID-/AIR-COOLED

Architecture scales with the thermal envelope

INTEGRATED SIGNAL

Interlock, presence and temperature options

TE AT EVERY CRITICAL INTERFACE

AC Whip • Shelf I/O • HV Busbar Connector • Rack-to-Rack • Vertical Busbar • IT Gear

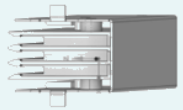
One Sidecar Architecture, Two Paths to the Compute Load

Sidecar Separates Power Conversion from Compute. Output Architecture Determines Where Final Conversion Occurs

PATH A

HIGH-VOLTAGE IT RACK

High Vol busbar connector



High Vol busbar

Rack-to-rack connections

PATH B

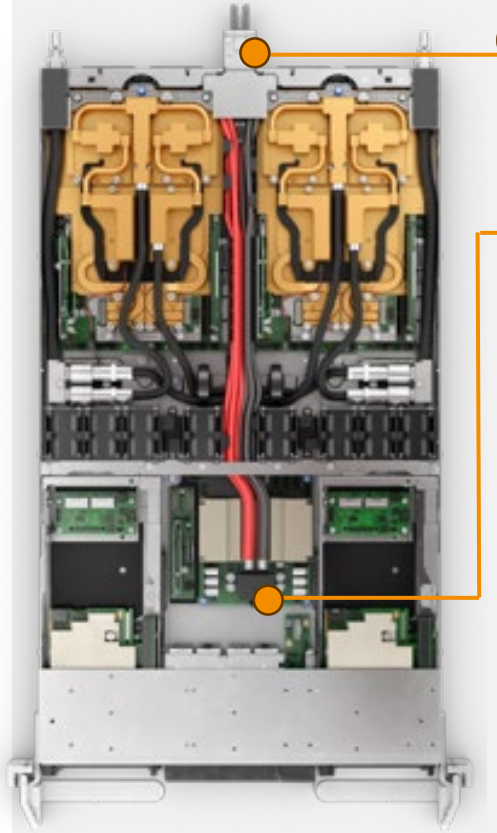
50V IT RACK

BB4000

Liquid-cooled busbar

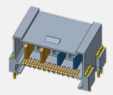
MULTI-BEAM HV to LV Connector

Inside the Compute Tray

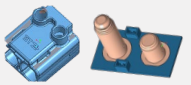


1 Power input via IT-Gear

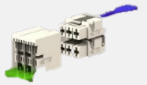
2 Terminating in the Tray



ELCON Micro+ Connector

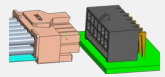


RAPID LOCK Connector



ELCON Mini connectors

ICCON Socket Insert



12/48V MPIC

IT Gear, ELCON Micro+, RAPID LOCK connectors & ICCON socket inserts available in 50V and High Voltage

TE AT EVERY CRITICAL INTERFACE AC Whip • Shelf I/O • HV Busbar Connector • Rack-to-Rack • Vertical Busbar • IT Gear

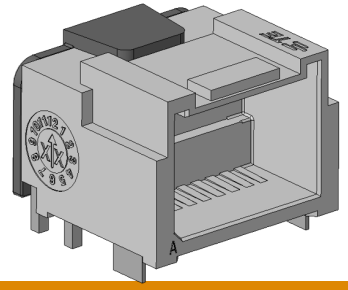
Helping You Design for Power Integrity, Safety and Sensing at the Interface

Clean Power

POWER INTEGRITY
Power density requires:
- More shielding
- Lower loop inductance
- B-field and voltage-drop control

LOWER INDUCTANCE
Higher current demands:
- Reduced voltage droop
- Low inductance and low resistance for the last-mile connectors

Improving compute performance



LEARN MORE AT TE PRESENTATION

"Reducing Parasitics, Improving Performance: The Case for Low Inductance"

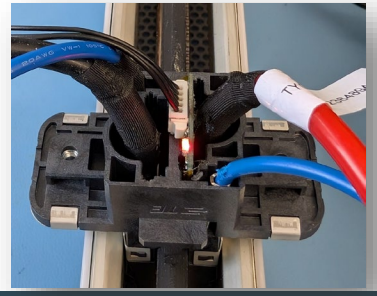
Tuesday, Oct 13
3:40 PM
SJCC – Concourse Level – 212

Smart Power

SENSING
Help anticipate failures by measuring:
- Temperature, presence, current and conditions

SAFETY
Enhance safety using:
- Touch-safe geometry, sequencing, interlock and de-energized service

Failures Become Predictable



See TE's Smart Power — TE Booth D3

AI Economics Are Reshaping The Interconnect

AI demands are driving larger compute fabrics …

AI Economics
More tokens per dollar

Larger Compute Domains
More GPUs/accelerators acting as one system

Higher Bandwidth / Radix
More data across more connections

… increasing the burden on interconnect

BANDWIDTH DENSITY
More lanes at higher speeds stressing electrical channels

REACH
Larger domains require longer high-bandwidth links

ENERGY / BIT
Equalization and retiming power rise with speed + distance

Architecture Evolution

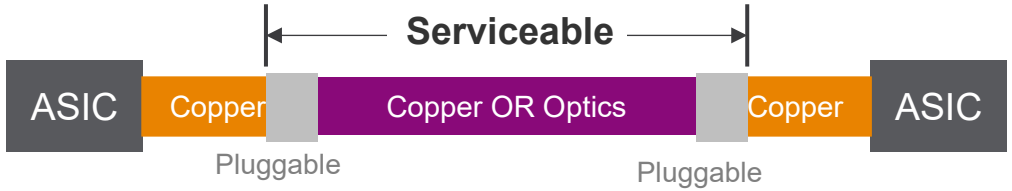
AI Scale is Pushing Electrical I/O to its Limits, Moving the Optical Boundary Closer to Compute

Moving Optics Inward Redefines The Service Model

Optical boundary is moving from faceplate to package

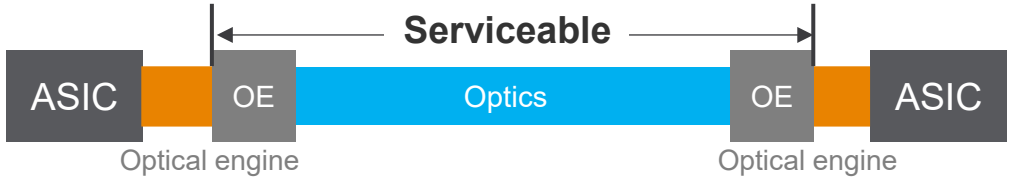
DAC through Pluggable Optics

I/O at faceplate



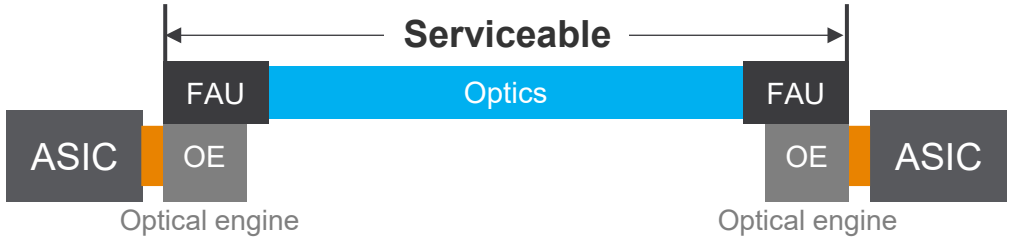
NPO

Optical I/O inside chassis



CPO

Optical I/O adjacent to ASIC



Serviceability must be engineered into the path

PROTECT HIGH-VALUE COMPUTE

Service optics and lasers without replacing the ASIC/package

CREATE SEPARABLE INTERFACES

Enable low-loss, repeatable optical disconnect and reconnect

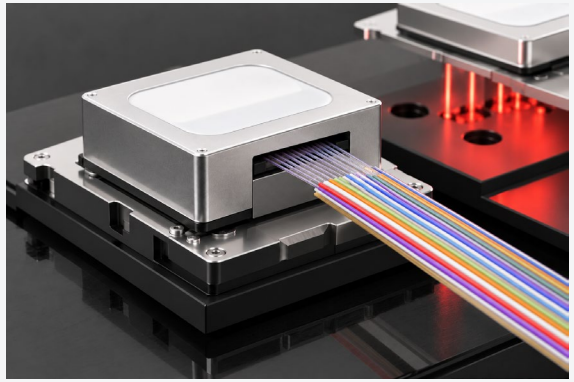
LOCK IN OPTICAL PRECISION

Establish alignment in manufacturing; enable passive, repeatable remating

As Optics Moves Inward, the Passive Optical Path becomes Integral to System Architecture

Scaling the Passive Optical Path – From Chip to System

CHIP ↔ FIBER
ATTACH



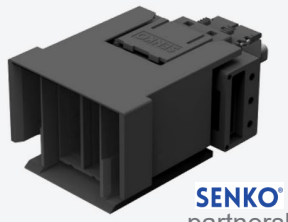
Illustrative

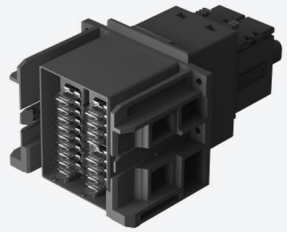
Fiber Array Unit (FAU)

Precisely couple high-density (including 2D M×N) fiber arrays to photonic engines

PRECISION • DENSITY • AUTOMATION

FIBER ↔ FIBER
CONNECT





SENKO partnership

USCONEC partnership

3M partnership

Connectors & Adapters

Create low-loss, separable optical interfaces for assembly, service, and upgrade

LOW LOSS • RELIABILITY • SERVICEABILITY

SYSTEM INTERCONNECT
ROUTE

Illustrative

Fanouts, Backplanes, & Fiber Assemblies

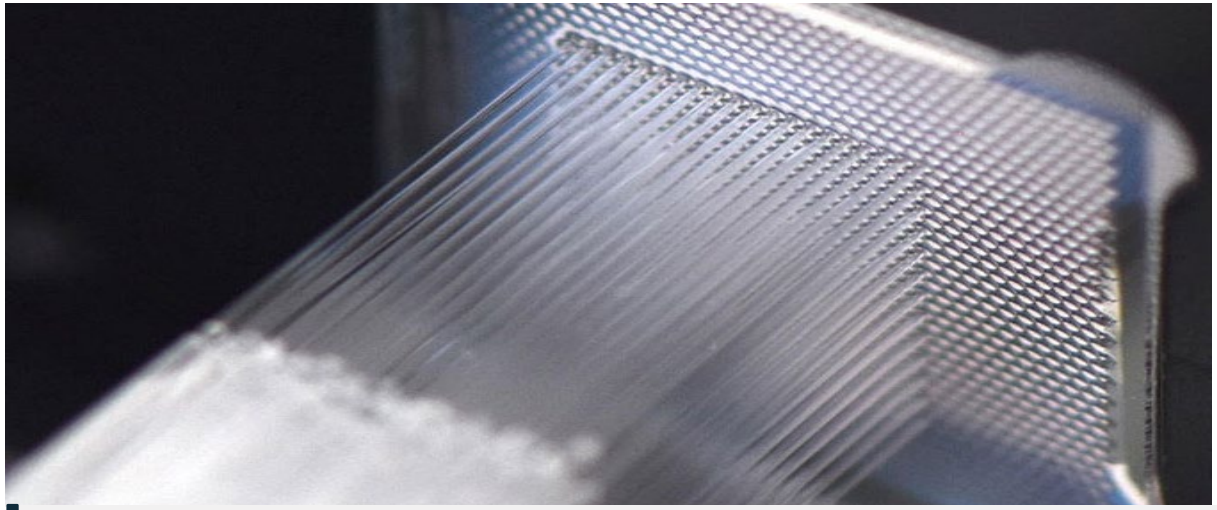
Distribute and manage high fiber counts across boards, racks and systems

ROUTING • DENSITY • SYSTEM INTEGRATION

FusionForge: Precision Fiber Termination At Scale

PRECISION ALIGNMENT

AUTOMATED LASER TERMINATION



HIGH-DENSITY FIBER ARRAY

HIGH DENSITY WITH GEOMETRY FREEDOM

- Flexible 1D & 2D array architectures
- Tight pitch and high channel counts
- Adaptable across photonic & coupling architectures

PERFORMANCE BY DESIGN

- Sub-micron precision and ±1deg PM clocking
- Epoxy-free optical interface
- Repeatable optical interface

AUTOMATION BUILT FOR SCALE

- Automated alignment and laser termination
- Repeatable process with higher throughput (vs. manual labor)

How TE is Engineering the Future of AI

PARTNER • SCALE • SUSTAIN

01 Ecosystem Co-Innovation

Accelerate innovation through ecosystem collaboration

- Ecosystem partnerships spanning second-sourcing, co-development, and standards influence
- Proven cross-licensing experience
- Committed design partnership from concept to ramp

02 Scalability for Growth

Translate innovation into repeatable execution

- Global footprint expansion
- Operations and engineering scalability
- High-performance product portfolios
- Committed to quality and rapid ramp

03 Sustainable Engineering

Design for performance, efficiency and lifecycle impact

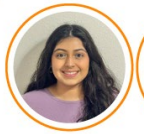
- Enhanced manufacturing engineering processes
- Product engineering sustainability – material, processes, lifecycle
- Energy efficiency across engineering and operations

JOIN US FOR EXPERT SESSIONS + SEE HOW WE'RE ENGINEERING THE FUTURE OF AI IN BOOTH D3

REDUCING PARASITICS, IMPROVING PERFORMANCE: THE CASE FOR LOW INDUCTANCE INTERCONNECTS

Tony Howard
Staff System Architect

OCT 13 | 3:40 PM



CO-PACKAGED COPPER AT 448G: BALANCING ELECTRICAL EFFICIENCY WITH SYSTEM LEVEL COMPLEXITY

Sakina Naqvi
System Architect

Usman Saeed
System Architect

OCT 14 | 9:05 AM

PANEL: 400G/LANE OPTICS MODULE CONNECTORS

Matthew Ho
Principal System Architect

OCT 14 | 11:05 AM

PANEL: STANDARDS AND FUTURE TECHNOLOGIES

Nathan Tracy
Senior Principal Engineer | OIF President

OCT 14 | 4:40 PM

ENABLING 1MW+ IT RACK POWER WITH AN OPEN RACK BUSBAR INTERFACE

Hal Loket
System Architect

Aniket Singh
Global Product Manager

OCT 15 | 10:15 AM

COOLING THE 448G FRONTIER: THERMAL ARCHITECTURE FOR 500 W-CLASS HSIO USING DIRECT-TO-CHIP LIQUID COOLING

Divya Bandaru
Staff System Architect

Sridhar Mukthineni
Sr. Manager R&D- Mechanical Center of Excellence

OCT 15 | 10:20 AM

ENGINEERING WHAT'S NEXT.

EVERY CONNECTION COUNTS

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